

Adam McMillan

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AI — Machine Learning — Psychology — Software Engineering — Music

Adam is a current Master of Information Technology student at the University of Melbourne, undertaking specialisation in Artificial Intelligence. He is currently working on a research project in ML fairness under the supervision of Jean Honorio. He has a keen interest in ML/AI, and his background in psychology provides him with a unique perspective, allowing him to solve problems with a scientific, yet human-centred approach. Outside of tech, Adam is a passionate classical pianist, performing and teaching regularly around Melbourne.

Technical Skills

- **Machine Learning:** Supervised learning (classification, regression, linear & non-linear models), unsupervised learning (clustering), fairness learning, semi-supervised learning (active learning), model evaluation (bias/variance tradeoff and generalisation), feature selection, ensemble learning, neural networks, generative models, anomaly detection
- **Statistical Analysis:** SPSS, Jamovi, R
- **Programming Languages:** Python, Django, JavaScript, React, C++, Java, Prolog, Haskell, GLSL
- **Libraries/Frameworks:** Scikit-learn, Pandas, Numpy, TensorFlow, Keras, PyTorch
- **AI Planning:** Search algorithms, heuristics, classical planning, reinforcement learning, game theory
- **Cloud/Cluster:** Kubernetes, Docker, Elastic, Fission, MPI, HPC, Google Cloud
- **Database Management:** SQL
- **Collaboration:** GitHub, GitLab

Experience

Software Developer Intern – Exergenics (July 2025 – Oct 2025)

Gained proficiency in data cleaning, modelling, evaluation, in the context of HVAC system optimisation using Exergenics' in-house tools. Contributed substantially to the development of new features and maintenance of Exergenics' software.

Staff Accompanist – University of Melbourne (2023 – Present)

Providing piano accompaniment for students of the Melbourne Conservatorium of Music (University of Melbourne) for recitals and performance classes.

Piano Instructor – Self (2011 – Present)

Private piano teaching, in my own studio as well as for schools such as Melbourne Grammar School, Parkville Music School, and Kensington Primary School. Supported many students in achieving competition and exam success.

Education

Master of Information Technology (Artificial Intelligence)

University of Melbourne

2024 – present (expected completion: late 2026)

Bachelor of Psychology (Honours)

Deakin University, Cognitive Neuroscience Unit

Class I Honours

2019 – 2023

Bachelor of Music (Honours)

Griffith University, Queensland Conservatorium of Music

Class I Honours

2011 – 2014

Awards

- Dean's Honours List, 2025 (University of Melbourne)
- Deakin Scholarship for Excellence (Deakin University)
- Sir Samuel Griffith Scholarship (Griffith University)
- University Medal (Griffith University)
- 1st Prize, Great Romantics Competition (Melbourne Recital Centre)
- 1st Prize, Queensland Piano Competition

Research

Novel algorithms for exploratory fair classifier learning

Masters research at University of Melbourne under supervision of Jean Honorio, (current)

Exploring the role of motor imagery in disordered movement in children with ADHD

Honours project at Cognitive Neuroscience Unit, Deakin University, 2023

Perfectionism in classical pianists: An exploration of how it manifests itself and possible coping strategies

Honours project at Queensland Conservatorium, Griffith University 2014